THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The current that changes direction periodically is 1

(a) alternating current **(b)** static current

(c) direct current **(d)** steady current

2. The magnetic field at a point due to a straight current-carrying wire 1

(a) decreases as the distance from the wire increases

(b) is zero everywhere

(c) increases as the distance increases

(d) is independent of distance

3. In Fleming's left-hand rule, the thumb points in the direction of 1

(a) the induced emf **(b)** the current

(c) the field **(d)** the force (motion)

4. The frequency of AC supplied in India is 1

(a) 60 Hz **(b)** 50 Hz **(c)** 25 Hz **(d)** 100 Hz

5. The magnetic field produced by a current-carrying wire at a given point depends on 1

(a) the colour of the wire

(b) the current and the distance from the wire

(c) the length of the wire only

(d) the voltage only

6. In Fleming's left-hand rule, the middle finger points in the direction of 1

(a) current **(b)** field **(c)** velocity **(d)** force

7. In a household circuit, the wire with red insulation is the 1

(a) fuse wire **(b)** earth wire **(c)** live wire **(d)** neutral wire

8. The magnetic field inside a long current-carrying solenoid is 1

(a) zero **(b)** increasing along the axis

(c) circular **(d)** nearly uniform

9. A positively charged particle moving perpendicular to a magnetic field experiences a force that is 1

(a) perpendicular to both **(b)** zero

(c) along the velocity **(d)** along the field

10. The earth wire is connected to 1

(a) the fuse **(b)** the neutral terminal

(c) the live terminal **(d)** the metal body of the appliance

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What do you mean by overloading? 2

12. Why does a compass needle get deflected near a current-carrying wire? 2

13. What is the function of an earth wire? 2

14. Name the two parts of a simple generator that cause the current to be taken out. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. List the factors on which the strength of the magnetic field of a solenoid depends. 3

16. What is the difference between AC and DC? 3

17. What is the direction of the field inside a solenoid when the current flows clockwise as seen from one end? State the rule you used. 3

OR

A current-carrying conductor is placed perpendicular to the magnetic field. What will happen if the direction of the current is reversed? 3

18. Why is AC preferred for power transmission? 3

OR

Mention two sources of overloading in a household circuit and the safety measures. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the construction and working of an AC generator with the principle. 5

OR

Explain how a current-carrying wire experiences a force in a magnetic field. How can you increase the force and reverse its direction? 5

20. Differentiate between AC and DC. State the frequency of AC in India and give one advantage of AC. 5

OR

Compare the principles and energy conversions of an electromagnet and a generator. 5

END OF QUESTION PAPER Total: 40 Marks

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